The Letter That Suffers

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Abstract

There is an economical brutality to Schopenhauer's conception of existence. Beneath the elaborate architecture of human motives, ambitions and explanations lies something simpler: striving without final destination. Desire succeeds desire because satisfaction cannot terminate the thing that desires. To live is to want; to want is to lack; and to lack is already to have admitted suffering.

Blind Purposeless [Striving], BPS: Difficult to improve upon as an indictment, but perhaps the final letter has been resolved too quickly. Long before there was anything capable of striving, well before suffering, and before anything capable of asking what either was *for*, there was already S. And that S had or embodied other ideas...

I. Stationarity

In mechanics we write

$$S[q] = \int L(q, \dot{q}, t)\, dt,$$

with the physical trajectory satisfying

$$\delta S = 0.$$

Here S is **Action**, although the name is unfortunate for our present purposes because Action is not action in the ordinary sense. It is not a deed: it carries units of energy multiplied by time and possesses no notion of intentionality whatsoever. Yet the mathematics has always invited such an intentional language: light takes a stationary optical path; a mechanical system follows a stationary action; nature behaves *as if* neighbouring possibilities had somehow been compared.

They have not. The photon does not inspect the paths ahead, and a stone does not calculate its fall. The counterfactual alternatives belong to our mathematical description not the deliberations of nature. A path may therefore look chosen without there having been a chooser: order does not require intention, and optimisation, or at least for mathematical appearances sake, does not require purpose. Our first S, then, is not Striving but **Stationarity**. For the moment Schopenhauer may keep both his adjectives, blind and purposeless.

II. States

Boltzmann gives us another S:

$$S = k_{\mathrm{B}} \ln \Omega,$$

where S is entropy and Ω counts the possible microscopic **States** compatible with the macroscopic condition we observe. The happenstance recurrence of the letter is mereley suggestive, coaxing us to follow through on a lineage. To specify an organised thing is, among other things, to exclude possibilities. A heap and a house might contain much the same material, yet what distinguishes the house is not some additional substance called *housemess*; it is the severe restriction placed upon where its components interrelate and may be .

A protein, likewise is not merely a collection of atoms, as a membrane is more than a collection of molecules, and a cell more than the sum of its ingredients stirred together. Organisation constrains states. Structure is expensive among the plethora of possibilities. Entropy and structure are therefore neither synonyms nor simple opposites, meeting in a bookkeeper's accounting of possible states. The universe, despite its tendency towards increasing entropy, has proved remarkably fertile in exceptional arrangements, and understanding how that comes to pass happens to require us to look closer at that which we call gravity .

III. Structure

Gravity complicates our nursery rhyme version of the Second Law in which entropy merely turns everything inexorably to mush. A nearly homogeneous gravitating universe is not already at gravitational equilibrium, it is merely nearly: small inhomogeneities grow, matter falls towards matter, and gravitational potential energy becomes kinetic energy and heat. Clouds contract, dense centres form, and some eventually ignite as stars. None of this requires intention. No to begotten star attracts its birthing gas cloud towards its destiny, no stellar blueprint is realised; inhomogeneities are necessary local dynamics acting upon suitable initial conditions are sufficient.

Yet those blind dynamics produce **Structure**, and a sufficiently particular structure can shine, establishing an enormous temperature difference between itself and the colder universe surrounding it. Blindness has not disappeared and purpose need not have entered, but the range of consequences has altered simply because matter has acquired form.

IV. The Shoulder of a Photon

A star exports energy as radiation, and a planet as our duly does may intercept a tiny fraction of that flux. The significant fact is that while energy arrives, approximately the same energy must eventually leave again; what matters is that change in **quality** of that energy. Radiation arriving from a hot source carries comparatively little entropy per unit energy, while a planet ultimately returns that energy towards colder surroundings principally as radiation, both at a lower characteristic temperature and with greater entropy. Between arrival and departure lies a gradient available to be *Spent*.

We come from a rich heritage of consumption and spending. For life, even that rare form susceptible to marketing occupies that gradient. Trace back that affintiy to an advertisement to a photon caught by a pigment molecule whose energy is converted into chemical free energy, to enter into carbohydrates, tissues, herbivores, predators, decomposers and the innumerable transactions through which living structures maintain themselves. Energy is degraded and entropy is produced, yet before the gradient is exhausted matter has had the opportunity to organise, repair, reproduce, sense and move. The Second Law not defied; as the local maintenance of intricate organisation accompanies an increase of entropy in the enveloping system of which we represent but a mote of dust. Life far from being an exemption from dissipation is the product of its profligacy.

There is something indecently consequential about the photon in this story. Beginning as it does as part of a blind stellar flux and ending, after many a transformation, to help to sustain a particular conglomeration of matter capable of intercepting another of its ilk with an eye.

That same bosonic light by which the living world is sustained becomes, eventually, the light by which some part of that world can see.

V. Self-maintenance

A stone may persist without doing anything in particular to remain a stone. A living structure needs to sustain through sustenance. Its organisation depends upon continued throughput: membranes maintain gradients, cells regulate concentrations, proteins are repaired or replaced, and organisms acquire matter and free energy while disposing of degraded products. A living structure must continually do work merely to remai recognisably itself. Structure has become **Self-maintenance**.

That changes the logical landscape because there are now possible states from which the structure can continue and possible states from which it cannot. There is still no requirement for consciousness, intention or foresight; a bacterium possesses no theory of tomorrow. Nevertheless, something consequential has been happened upon that would have been meaningless in worlds of just photon or stone: one outcome can be better *for it* than another. Not morally better or cosmically preferred, but functionally better. There is now something that can fail, and therefore something relative to which success can acquire a physical meaning.

VI. Selection

Living structures through differential binding dynamics have differentiated, *individuated* no less and some of those differences are adaptions to alter their capacity to acquire energy, escape destruction, repair damage or reproduce. Such differences accumulate without imagining, no formulation of an optimisation problem, no future creature reaching backwards through time to summon the anatomy it will require. **Selection** remains blind, yet what survives selection can possess function.

The wing is for flying as a falling stone is not for reaching the ground, and the eye is for seeing in a sense in which the Sun is not for illuminating the eye. These statements can and should be translated into causal histories, but doing so does not abolish the function; it explains how function became possible without foresight. Blind processes have produced structures whose present amalgamation can be understood partly in terms of what that organisation accomplishes.

Here, perhaps, is the first serious wound to Schopenhauer's **Purposeless**. Nothing supernatural has entered the story, and no purpose has been imposed upon nature from outside it. Nevertheless, a thing has acquired a *how is it so* without the universe having acquired a *for what reason.*

VII. Survival

Selection leaves behind structures equipped, with varying success, to continue, and continuation itself now acquires a peculiar status. For a stone, persistence is simply what happens until something else happens. For an organism, persistence is accomplished through feeding, regulation, repair, escape and reproduction. These activities differ enormously, but each participates in maintaining an organisation that would otherwise decay. Such is **Survival**.

Nothing yet requires a creature to know that it is surviving. Nevertheless, something now exists for which a future state can objectively constitute success or failure. The universe need have no purpose for the organism, but the organism has acquired something remarkably close to a purpose within the universe: there are things it must do if it is to continue being the thing it is. The distinction between cause and purpose remains important, but it no longer follows that because the former explains the latter, the latter must be unreal.

VIII. Seeking

Once a living structure can sense differences in its environment and alter its behaviour accordingly, Survival becomes **Seeking**. A plant grows towards light, a bacterium follows a nutrient gradient, an animal searches for food, a predator pursues and prey escapes. Each stands downstream of stellar free energy, shouldering through another transformation the ancient gradient between a hot star and cold space.

Purposive language becomes progressively harder to avoid. The gazelle runs to escape, the lion hunts to eat, and the bird builds to nest. There is no difficulty supplying mechanisms beneath these sentences: muscles contract because motor neurons fire, motor neurons fire because sensory systems respond, and those systems exist because ancestral differences affected subsequent reproduction. Nothing in that explanation requires cosmic purpose, yet neither does it make the gazelle's escape fictitious. At some scale, what happens next has begun to matter to the thing to which it happens. The world contains more than mere projectiles following trajectories but organisms suffering consequences.

Seeking is therefore more than movement, being movement rendered asymmetric by the continued existence of the mover. Once a system detects the difference between what sustains it and what threatens it, the beginnings of an apparent teleology are no longer supplied only by our description. They have entered the organisation of the system itself.

IX. Striving

A structure that must continually maintain itself cannot merely persist passively: it must act, and a structure capable of detecting threats to that maintenance must respond. Once it can also represent possible futures, it can do something more remarkable still, acting in the present because of a state that does not yet exist. There is now the state that is and the state that might be, and between them lies desire.

At this point Schopenhauer becomes harder rather than easier to dismiss. He placed Striving beneath the phenomenal world as its underlying character, and nothing in the lineage developed here disproves that claim. He might reasonably reply that every star, cell, appetite and thought described above is simply another objectification of the *Will*. The more modest claim available to us is only that what we experience as striving need not be inserted into the beginning of the physical story in order to become undeniably present within it. Structure can precede Striving, even if that historical precedence cannot settle what Striving ultimately is.

Something important has nevertheless happened. The earlier organism could be said to have states that were better or worse for it; the striving organism can in some measure represent those states before they arrive. A future that does not yet exist can thereby become causally consequential in the present, because a present organism carries within itself a model, appetite, expectation or fear concerning what may come next. Blind nature has produced something capable of anticipation.

X. Subjectivity

There is another distinction that becomes difficult only very late in this story: the distinction between *object* and *subject*. For almost all of the lineage there are objects in the ordinary physical sense. Matter falls, stars burn, photons propagate, molecules collide and structures form and decay. Things happen. With life there appear objects for which what happens has consequences, but eventually there appears something stranger still: an object becomes capable of experiencing what happens to it. The object becomes a **Subject**.

Nothing in that transition exempts it from objecthood. The subject still has mass and exchanges energy; its neurons obey electrochemical constraints, its tissues age, and its organisation must continually be purchased against equilibration. We do not cease to be objects when we become subjects. We become objects that possess a point of view, and this may be the extraordinary privilege of coming into being: one gets a world, not merely a world that exists but a world that *appears*.

Warmth is no longer molecular motion alone because there is also feeling warm; light is no longer electromagnetic radiation alone because there is brightness; injury is no longer damaged tissue alone because there is pain; and a past configuration of matter may be not merely absent but lost. Objective reality has acquired subjective reality without surrendering an atom of its objectivity. There is a state of the world and there is what that state is like for the creature within it. The first may be measured from outside; the second has an inside. Both are real.

The photon striking a retina belongs to objective reality, while its apparent redness is the purview of subjective reality. A damaged nerve can be described objectively: pain is what damage can become when there is someone for whom the damage occurs. Consciousness may therefore be less a window placed upon reality than a peculiar doubling of it. The universe continues to happen and now, somewhere within the happening, it also seems. Some of its objects have *come to their senses*.

XI. Subjection

The word *subject*, however, carries its own warning. We become subjects without ceasing to be objects, and remain **Subjected** to all the conditions from which subjectivity arose: to gravity and entropy, to hunger and time, to the requirements of our own maintenance, and eventually to other subjects. To acquire a point of view is also to acquire the possibility of occupying somebody else's.

One can become an object of attention, affection, appetite or desire, and this may be among the stranger humiliations contained within the privilege of subjectivity: an object becomes a subject only to discover that another subject can make an object of it again. At the crudest biological resolution, reproduction contains this recursion particularly starkly. Subjects become objects of desire to other subjects and may thereby participate in bringing another subject into being. Selection has no embarrassment about this. Whatever other purposes conscious creatures subsequently discover for themselves, the lineage that produced them was filtered through reproduction. Subjects beget subjects.

There are bodies for whom reproduction is impossible, bodies for whom it is unwanted, and lives in which what was once imagined becomes unavailable; none thereby becomes less a subject. Yet we speak remarkably easily as though a life had somehow failed to complete its sentence when it does not reproduce. Perhaps that is Selection speaking through us, or perhaps that is another excuse.

Subjectivity therefore confers no immunity from the objective world. Quite the reverse: it makes possible a new manner of being exposed to it. We are subjects because there is something it is like to be us, and subjected because nothing about having an inside releases us from what can happen outside it. The privilege of perception arrives already attached to vulnerability. We are subjected to reality before we are permitted a perception of it.

XII. Suffering

And finally there is **Suffering**.

A stone cannot lament the state into which it falls. A conscious creature can do something worse than experience the state it occupies: it can represent another. It can compare what is with what might be, what was, or what can never be again.

Hunger represents what might satisfy it.Fear represents what might happen. Grief represents what cannot happen again. There are other examples. The important point is that suffering appears late. It requires an inside and, in its more elaborate forms, memory, counterfactuals, and the capacity to imagine a continuation different from the one that actually occurred. Perhaps suffering is the price paid by matter for acquiring the counterfactual. Glib perhaps for a price implies that somebody was offered the transaction. No one was.

There was no bargain. No consent to the terms.

The same physical world that produces an injury eventually produces something for which injury can hurt. The same passage of time that changes objective states eventually produces something capable of wishing that one of those states had remained. A photon does not wish that it had taken another path. We do. And tragically no amount of wishing varies the path already taken.

XIII. The Conscious Conceit

There is something extraordinary about where the lineage has arrived. A photon emitted by a star enters a chain of transformations that can eventually sustain a structure capable of seeing the star from which such photons come. Matter acquires eyes, then memory, then some representation of itself and of possible worlds beyond its present state. Eventually a portion of the universe becomes capable of asking what the universe is for.

There may be a conceit hidden in that question, perhaps even a deceit. Conscious creatures discover purpose everywhere partly because they are themselves products of systems in which distinctions between possible futures acquired consequences. We are optimisation engines surprised to discover optimisation, purpose-generating structures asking whether the structure that generated us had a purpose. We should therefore be suspicious of the ease with which the universe begins to resemble the kind of creature looking at it, but that suspicion does not make purpose unreal.

Reality routinely acquires legitimate descriptions at different scales. Money has no fundamental particle, yet debts are real; a boundary has no mass, yet a cell dies when its membrane fails; temperature is not a property we sensibly assign to an isolated molecule. Purpose might belong to this same class of realities: absent from the inventory of fundamental objects without therefore being absent from the world those objects eventually compose.

The subject, on this account, is neither outside objective reality nor merely carried passively within it. It remains an object upon which the world acts, but becomes an object through which the world can subsequently act differently. What is perceived changes what is sought, what is sought changes what is done, and what is done changes what subsequently exists. Once matter can represent alternatives, an unrealised future can become causally consequential through the present structure that represents it.

Blindness has therefore produced something capable of looking, while purposelessness has produced something capable of purposes. Whether that is an answer to Schopenhauer or merely another manifestation of his Will is still moot. It is enough for the present argument that purpose need not have been present at the beginning in order to become real somewhere along the way.

XIV. The Letter That Suffers

Only now can the final letter be allowed to show all the meanings it has accumulated. The sequence is merely an alliteration: Stationarity does not cause States because both begin with S; entropy is not Structure, Structure is not Selection, and Subjectivity is not Suffering. The recurrence of the letter is a conceit. Beneath it, however, lies something suggestive of an historical dependency as the letter has proliferated:

$$\boxed{\begin{aligned} S:\quad & \text{Stationarity} \longrightarrow \text{States} \longrightarrow \text{Structure} \\ & \longrightarrow \text{Self-maintenance} \longrightarrow \text{Selection} \longrightarrow \text{Survival} \\ & \longrightarrow \text{Seeking} \longrightarrow \text{Striving} \longrightarrow \text{Subjectivity} \\ & \longrightarrow \text{Subjection} \longrightarrow \text{Suffering.} \end{aligned}}$$

There could be no biological function before there were structures whose differences affected persistence, no Seeking before there were systems capable of detecting differences relevant to their continuation, and no conscious Striving before there were creatures capable of representing states that had not yet occurred. Subjectivity required some object to acquire an inside; Subjection in its human sense required one subject to become an object within the world of another; and Suffering over what might have been required something capable of imagining otherwise.

This returns us to BPS and to the letter that has become less secure as **S** has accumulated meanings. The unstable letter may finally be the one in the middle: **P**, Purposeless. The question is no longer whether fundamental physics contains purpose, but at what scale the assertion of purposelessness is intended to hold. At the level of fundamental dynamics it may hold entirely, and at the level of a star almost certainly; at the level of a bacterium moving towards glucose the answer becomes less simple, at the level of an animal choosing between possible futures less simple again, and at the level of a conscious creature capable of imagining consequences before acting the word acquires another difficulty.

Purpose need not have preceded the universe to become possible within it. There were once no subjects, grief or Schopenhauer. The universe has acquired predicates it did not possess at every earlier stage, and purpose might be one of them. To insist that purpose can be real only if it was present from the beginning would be to apply to it a condition we do not apply to life, consciousness, temperature or stars.

BPS need not therefore be answered by inventing a purpose for the universe. It may be enough that Blind Purposeless Structure eventually produced structures for which purposes became real. The privilege of coming into being is not that we escape the blind universe but that, briefly and locally, the blind universe is no longer blind everywhere: some of its objects open their eyes. Those objects remain subjected to the same reality from which they arose, yet they can perceive alternatives within it, prefer among them, and act in ways that alter what subsequently becomes real.

If conscious creatures are distinguished by anything, perhaps it is this: the trajectory before them need no longer be accepted merely because it is the trajectory upon which they arrived. There remains, while there remains anything, the possibility of a variant Action,

$$\boxed{\delta S}$$